PROJECT NAME:

"Comparative Bioinformatics Analysis of Insulin Protein Sequences Across Selected Mammalian Species"

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ABSTRACT:

Insulin is an important hormone that helps regulate blood glucose levels in mammals. Although insulin performs a similar function across species, differences in its protein sequence can provide useful information about its conservation and evolutionary relationships. The present study aimed to compare insulin protein sequences from ten selected mammalian species using bioinformatics tools. The sequences were collected using their accession numbers and analyzed for amino acid composition, physicochemical properties, conserved and variable sites, evolutionary distances, and phylogenetic relationships. Multiple sequence alignment and phylogenetic tree construction were performed using MEGA software, while ProtParam was used to examine the physicochemical properties of the proteins. R software was used to generate plots and perform statistical analyses.

The alignment contained 110 positions, of which 68 were identified as conserved and 42 as variable. Leucine was the most abundant amino acid, with an average composition of 17.38%. The molecular weights of the analyzed proteins ranged from 11,256.09 to 12,364.33 Da, and their theoretical isoelectric points ranged from 5.20 to 6.70. Pairwise evolutionary distances ranged from 0.0183 to 0.2472, indicating differences in sequence similarity among the selected species. Phylogenetic relationships were examined using both Neighbor Joining and Maximum Likelihood methods. Statistical analysis showed a positive correlation between the aliphatic index and GRAVY values ($r = 0.741$, $p = 0.01423$), whereas the correlation between molecular weight and GRAVY was not statistically significant ($r = -0.315$, $p = 0.3754$).

Overall, the findings show that mammalian insulin sequences share many conserved amino acid positions while also exhibiting variation in sequence composition and physicochemical properties. These results demonstrate the usefulness of comparative bioinformatics in studying protein conservation and evolutionary relationships among mammals.

KEYWORDS: Insulin, comparative bioinformatics, mammals, amino acid composition, conserved sites, evolutionary distance, phylogenetic analysis.

1. INTRODUCTION

Insulin is a peptide hormone secreted by pancreatic β-cells that plays an essential role in regulating blood glucose levels and maintaining normal energy metabolism. It regulates carbohydrate, lipid, and protein metabolism by promoting glucose uptake and storage while inhibiting glucose production, lipolysis, and protein degradation. Research on insulin and its analogues has contributed to a better understanding of its biological functions and therapeutic importance in diabetes management (Francis *et al*., 2024).

Although insulin performs an essential biological function, its gene sequence and processing have changed during mammalian evolution. Comparing insulin sequences from different species can help identify conserved

regions and evolutionary differences (Irwin, 2021). Insulin also influences the metabolism of fats and proteins. Because of its important biological functions, insulin has been widely studied in different organisms.

Insulin is found in many mammalian species, including humans, chimpanzees, mice, rats, cattle, pigs, horses, dogs, rabbits, and rhesus macaques. Although insulin performs a similar function in these species, differences in their amino acid sequences may occur. Comparative analysis of insulin and related proteins across organisms provides information about their evolutionary history and the conservation of important biological features. Studying insulin sequences from selected mammals can therefore help characterize similarities and differences among these species (Chrudinová *et al*., 2026).

Bioinformatics provides useful methods for comparing protein sequences and investigating their biological properties. Multiple sequence alignment can identify conserved and variable amino acid positions, while pairwise evolutionary distance analysis measures differences between sequences. Phylogenetic trees can also be constructed to examine patterns of similarity and evolutionary relationships among the selected organisms.

The physicochemical properties of proteins provide additional information about their composition and characteristics. Parameters such as molecular weight, theoretical isoelectric point, instability index, aliphatic index, and grand average hydropathicity (GRAVY) can be calculated using tools such as ProtParam. Amino acid composition analysis can further reveal which residues are common or uncommon in the selected proteins.

In the present study, insulin protein sequences from ten selected mammalian species were compared using bioinformatics approaches. MEGA software was used for multiple sequence alignment, conserved and variable site analysis, evolutionary distance estimation, and phylogenetic tree construction. ProtParam was used to investigate physicochemical properties, while R software was used to visualize the results and examine relationships between selected protein properties.

This comparative study aims to provide a clearer understanding of insulin sequence conservation, species specific variation, physicochemical characteristics, and evolutionary relationships among the selected mammals.

2. RESEARCH OBJECTIVES

The main objective of this study is to compare insulin protein sequences across ten selected mammalian species using bioinformatics tools.

The specific objectives are:

1. To collect insulin protein sequences from ten selected mammalian species using their accession numbers.
2. To analyze and compare the amino acid composition of the selected insulin proteins.
3. To determine the physicochemical properties of the proteins, including molecular weight, theoretical isoelectric point, instability index, aliphatic index, and GRAVY.
4. To identify conserved and variable amino acid positions through multiple sequence alignment.

5. To calculate pairwise evolutionary distances and compare sequence similarities among the selected species.
6. To construct Neighbor Joining and Maximum Likelihood phylogenetic trees to investigate evolutionary relationships.
7. To visualize the results using R software and examine correlations between selected physicochemical properties.
8. To interpret the combined findings to understand the conservation and variation of insulin proteins across the selected mammals.

3. MATERIALS AND METHODS

1.STUDY DESIGN

This study used a comparative bioinformatics approach to investigate insulin protein sequences from ten selected mammalian species. The analysis focused on amino acid composition, physicochemical properties, sequence conservation, evolutionary distances, and phylogenetic relationships.

2.PROTEIN SEQUENCE COLLECTION

Protein sequences were obtained from the National Center for Biotechnology Information (NCBI) Protein database. Ten insulin protein records representing selected mammalian species were chosen for comparative analysis. The accession number and organism name of each record were documented to maintain traceability and facilitate retrieval of the same sequences for future studies. The selected sequences were subsequently used for alignment, physicochemical characterization, and evolutionary analysis (National Center for Biotechnology Information [NCBI], n.d.).

3.PROTEIN SEQUENCE RETRIEVAL

No.	Scientific Name	Common Name	Accession Number
1	Homo sapiens	Human	AAA59172.1
2	Pan troglodytes	Chimpanzee	CAA43403.1
3	Macaca mulatta	Rhesus macaque	XP_028688882.1
4	Mus musculus	Mouse	NP_032413.1
5	Rattus norvegicus	Rat	CAA24560.1
6	Bos taurus	Cow	NP_776351.2
7	Sus scrofa	Pig	AAQ00990.1
8	Equus caballus	Horse	NP_001421663.1
9	Canis lupus familiar's	Dog	P01321.1
10	Oryctolagus cuniculus	Rabbit	AAA19033.1

Ten insulin protein sequences representing selected mammalian species were collected from public protein sequence databases. The accession numbers and organism names were recorded to ensure sequence traceability. The protein sequences were downloaded in FASTA format and organized for subsequent multiple sequence alignment, conserved and variable site analysis, pairwise evolutionary distance calculation, and phylogenetic tree construction. The same sequence records were used throughout the analysis to maintain consistency between the sequence dataset and the reported results.

4.AMINO ACID COMPOSITION ANALYSIS

The amino acid composition of each protein sequence was examined to determine the percentage of each amino acid. The results were organized in Microsoft Excel, and average amino acid percentages were calculated to compare residue composition across the ten sequences.

5.PHYSICOCHEMICAL PROPERTIES

The physicochemical properties of the selected insulin protein sequences were analyzed using ProtParam, an online tool available through the ExPASy server. The analysis included molecular weight, theoretical isoelectric point (pI), instability index, aliphatic index, and grand average of hydropathicity (GRAVY). These parameters were used to compare the predicted physicochemical characteristics of insulin proteins among the selected mammalian species. The results were recorded in a spreadsheet for comparison and graphical representation. ProtParam estimates these properties from the submitted amino acid sequences (Gasteiger et al., 2005).

6.MULTIPLE SEQUENCE ALIGNMENT

Multiple sequence alignment and evolutionary analyses were performed using Molecular Evolutionary Genetics Analysis (MEGA) 11. The alignment was examined to identify conserved and variable amino acid positions among the selected insulin protein sequences. Pairwise evolutionary distances were estimated using the Poisson amino acid substitution model with uniform rates and pairwise deletion, according to the settings used in the analysis. Phylogenetic relationships were investigated using Neighbor-Joining and Maximum Likelihood methods, with bootstrap analysis based on 1,000 replications to assess the support for tree branches. The resulting alignments, distance estimates, and phylogenetic trees were used to compare sequence similarity and evolutionary relationships among the selected mammals (Tamura et al., 2021).

7.CONSERVED AND VARIABLE SITE ANALYSIS

The aligned sequences were examined to identify conserved and variable amino acid positions. Conserved sites were positions showing the same amino acid across the analyzed sequences, while variable sites contained amino acid differences or gaps. The results were recorded for further interpretation.

8.EVOLUTIONARY DISTANCE ANALYSIS

Pairwise evolutionary distances were calculated in MEGA using the Poisson amino acid substitution model, uniform rates among sites, homogeneous lineages, and pairwise deletion of gaps and missing data. The resulting distance matrix was used to compare the degree of sequence divergence among the selected proteins.

9.PHYLOGENETIC ANALYSIS

Neighbor Joining (NJ) and Maximum Likelihood (ML) methods were used to investigate evolutionary relationships among the ten insulin protein sequences. Bootstrap analysis with 1,000 replications was performed to assess support for branches in the phylogenetic trees. The resulting trees were saved for comparison and interpretation.

10.STATISTICAL ANALYSIS AND VISUALIZATION

Statistical analysis and graphical representation were carried out using R in the RStudio environment. The physicochemical data collected for the ten insulin protein sequences were organized and analyzed to investigate patterns and relationships among the measured properties. Pearson correlation analysis was used to examine the relationship between aliphatic index and GRAVY, as well as the relationship between molecular weight and GRAVY. Correlation coefficients and p values were examined, using a significance threshold of $p < 0.05$. Graphs were prepared to present the distribution of selected physicochemical properties and to illustrate the relationships identified in the analysis. These statistical and graphical approaches supported the interpretation of variation among the selected mammalian insulin sequences.

11.DATA ORGANIZATION

The results were organized in Microsoft Excel into separate worksheets covering sequence information, amino acid composition, physicochemical properties, conserved sites, variable sites, evolutionary distances, phylogenetic analyses, and result interpretation. Figures and analysis files were retained as supporting project materials.

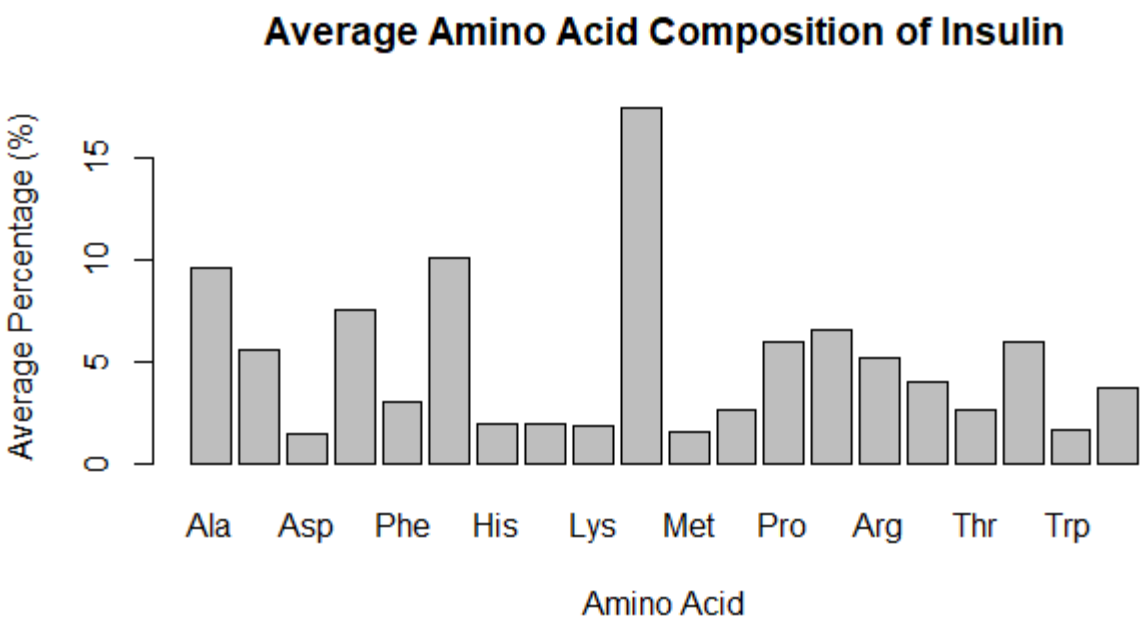
4.RESULTS

1.SEQUENCE INFORMATION

Ten insulin protein sequences from selected mammalian species were analyzed in this study. The sequences represented human, chimpanzee, rhesus macaque, mouse, rat, cow, pig, horse, dog, and rabbit. The sequence lengths ranged from 105 to 110 amino acids. Most of the selected sequences contained 110 amino acids, while the cow and pig sequences contained 105 and 108 amino acids, respectively.

2.AMINO ACID COMPOSITION

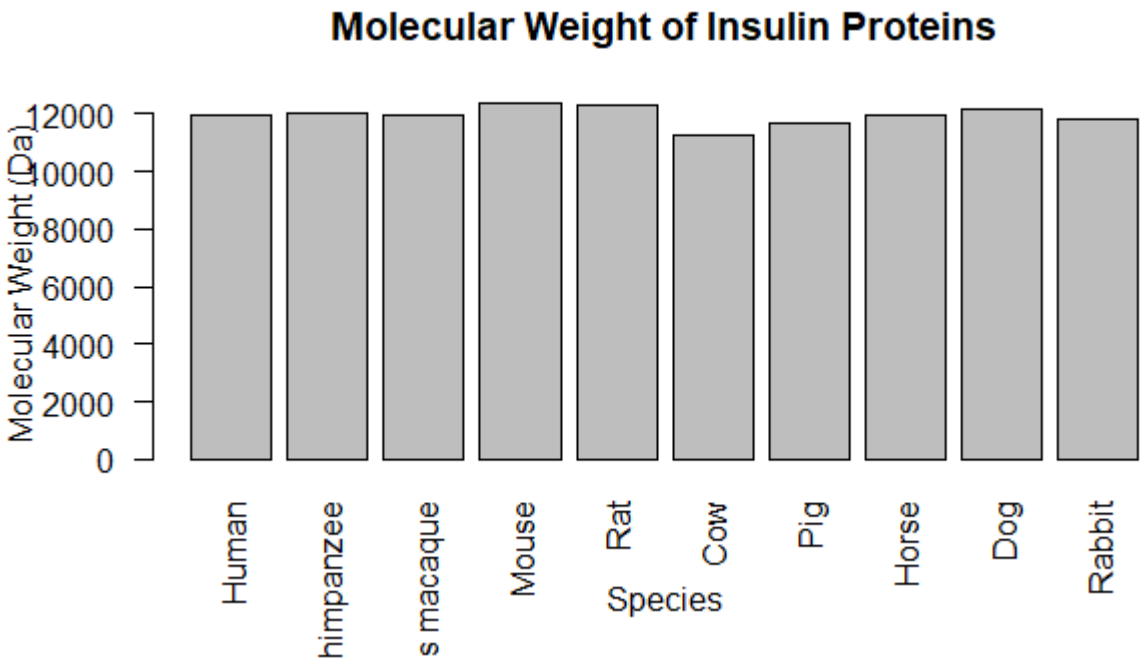
Amino acid composition analysis showed that leucine was the most abundant amino acid, with an average composition of 17.38%. Glycine was the second most abundant at 10.06%, followed by Alanine at 9.61%. The average sequence length was 109.3 amino acids. These results demonstrate differences in amino acid composition among the selected insulin proteins.



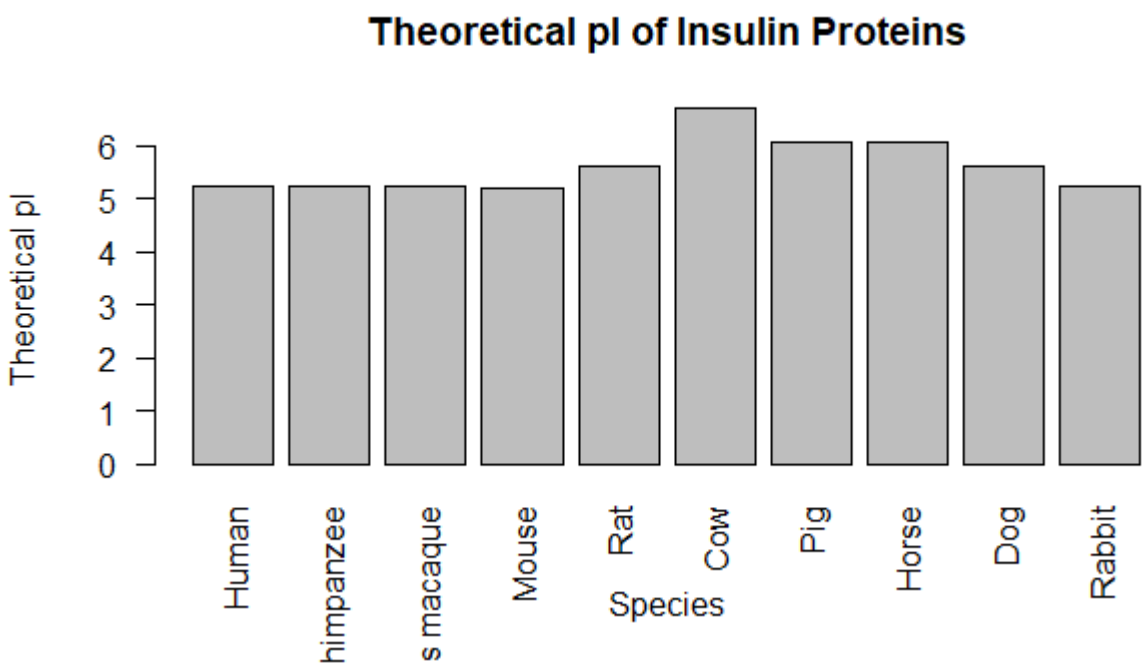
3.PHYSICOCHEMICAL PROPERTIES

The molecular weights of the analyzed proteins ranged from 11,256.09 to 12,364.33 Da. The theoretical isoelectric points ranged from 5.20 to 6.70. The instability index ranged from 33.91 to 48.54, while the aliphatic index ranged from 95.81 to 110.00. GRAVY values ranged from 0.106 to 0.289. These results indicate variation in the predicted physicochemical properties of the selected sequences.

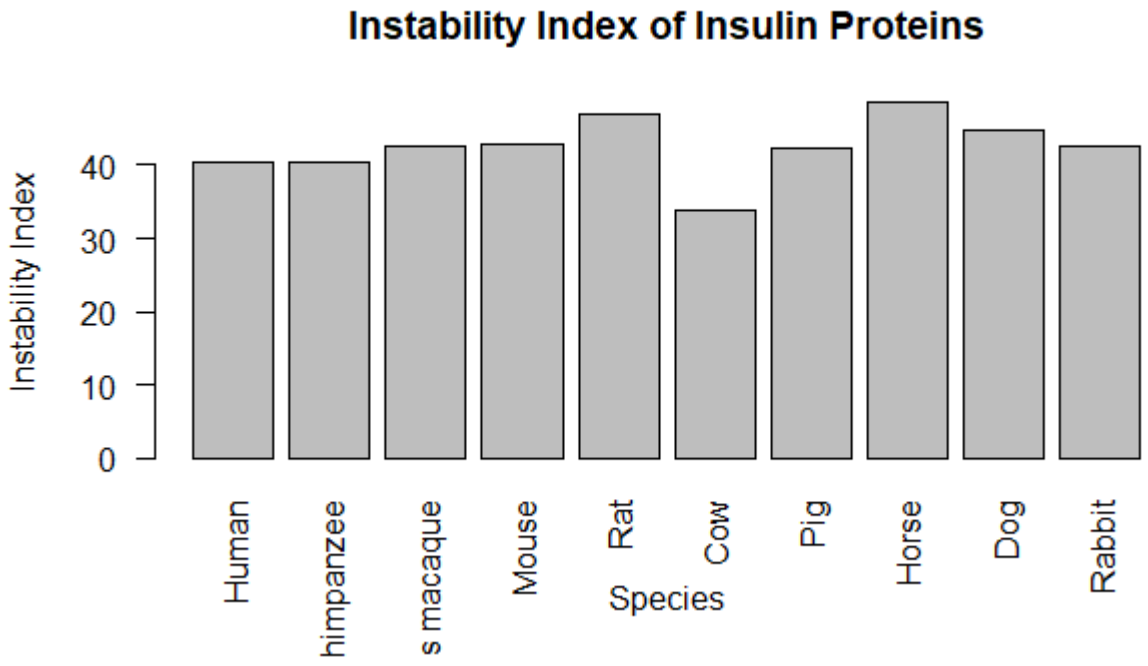
MOLECULAR WEIGHT GRAPH



THEORETICAL PI GRAPH



INSTABILITY INDEX

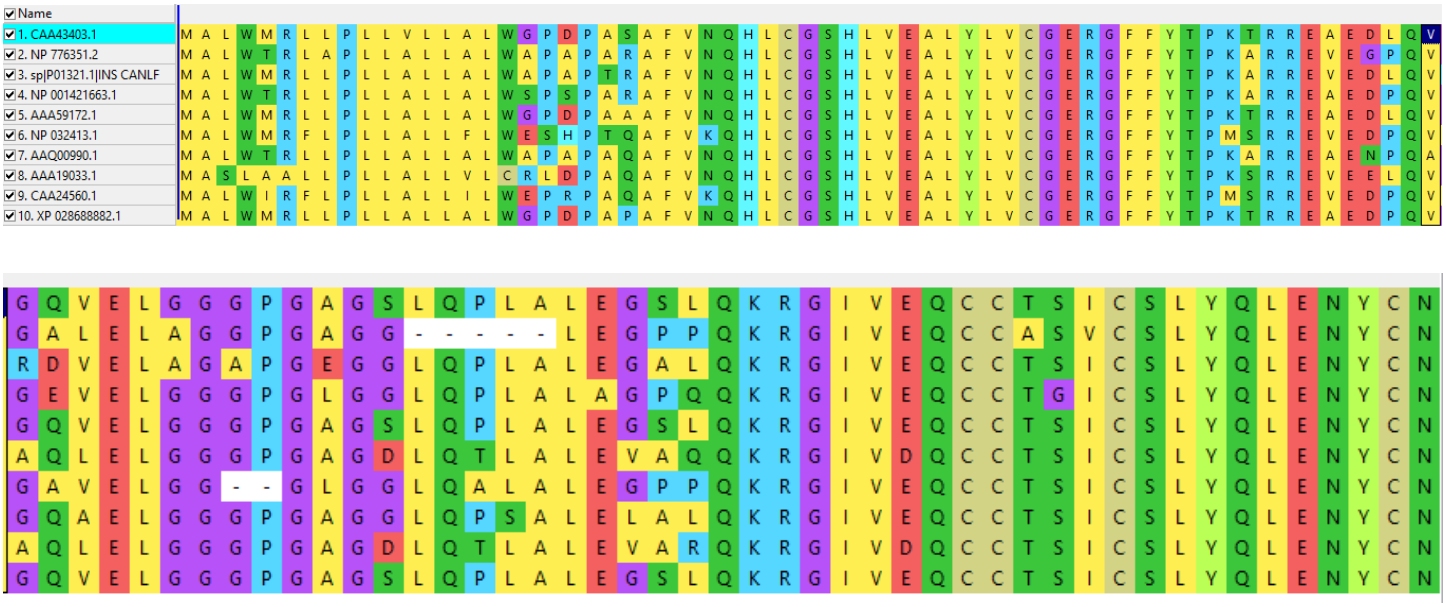


4.CONSERVED AND VARIABLE SITES

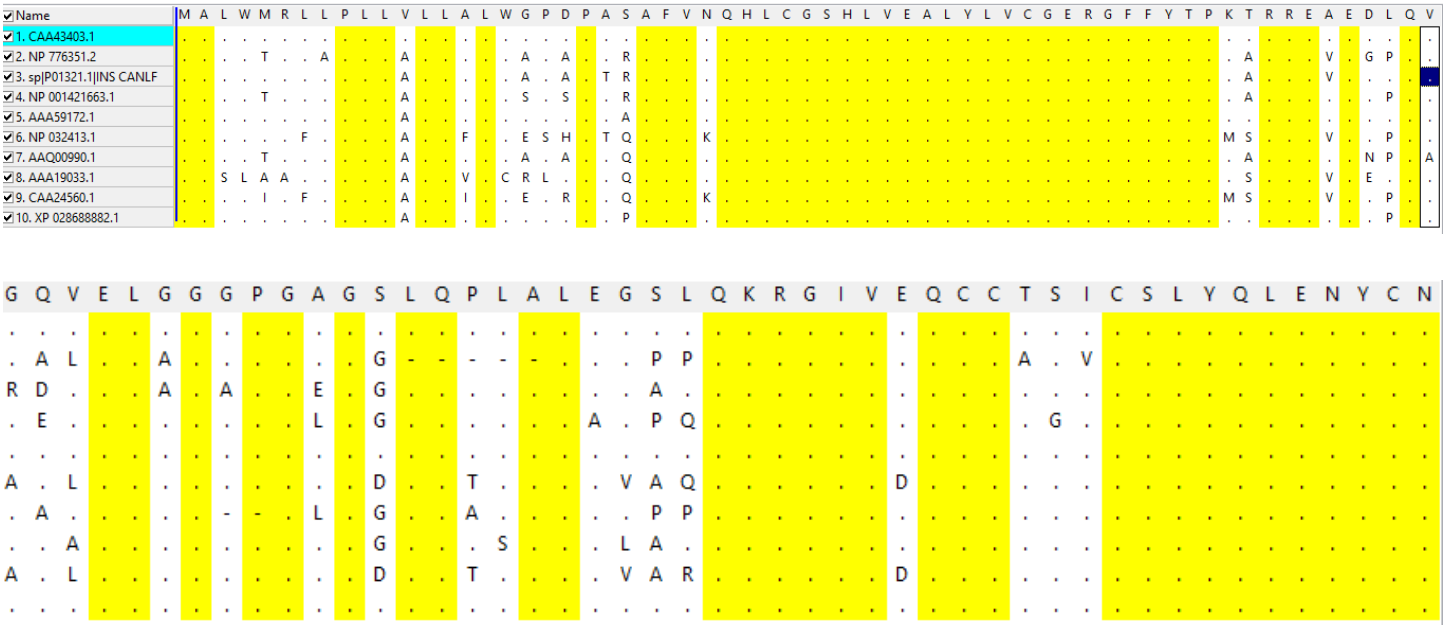
Multiple sequence alignment was used to investigate conserved and variable amino acid positions. The alignment contained 110 positions, with 68 identified as conserved and 42 as variable in the recorded analysis.

These findings indicate that the sequences share many amino acid positions while also containing differences among species.

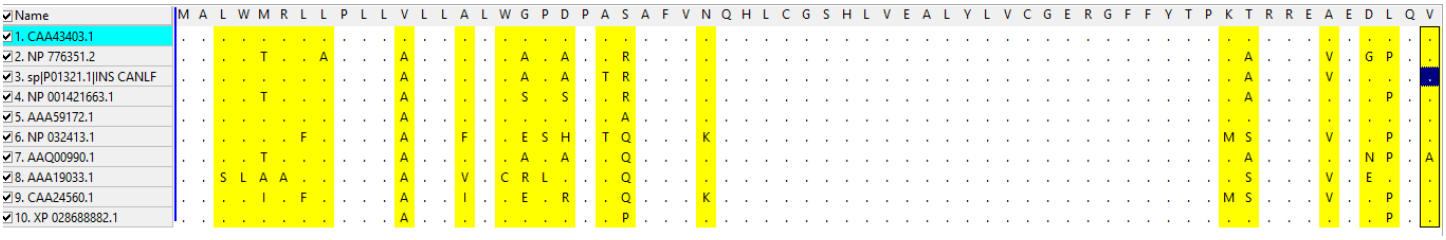
MULTIPLE SEQUENCE ALIGNMENT (MSA) OF 10 SPECIES

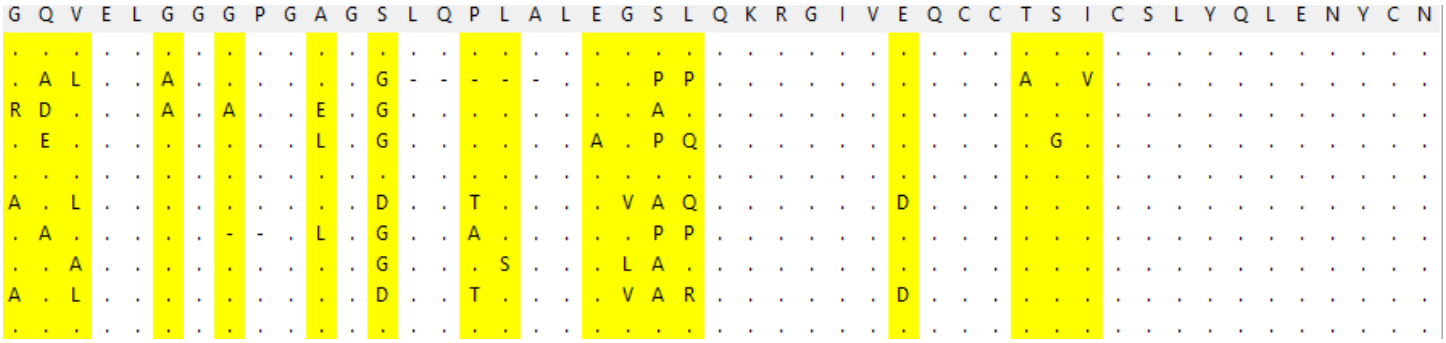


CONSERVED SITES



VARIABLE SITES

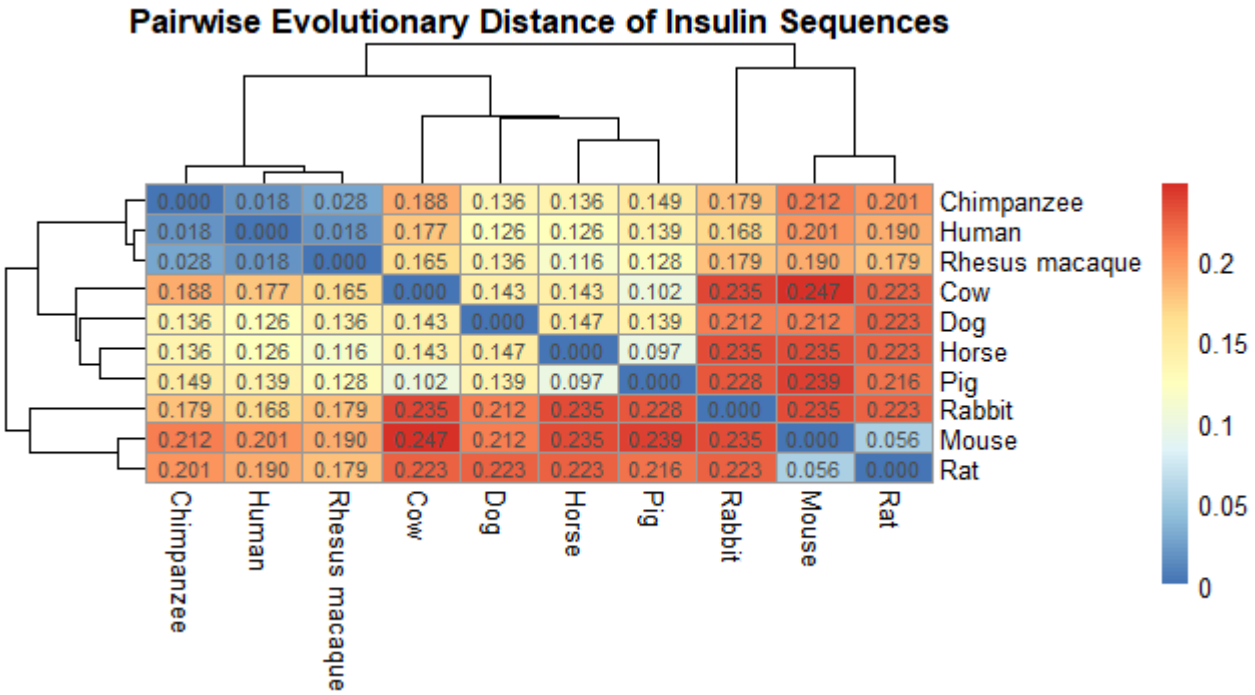




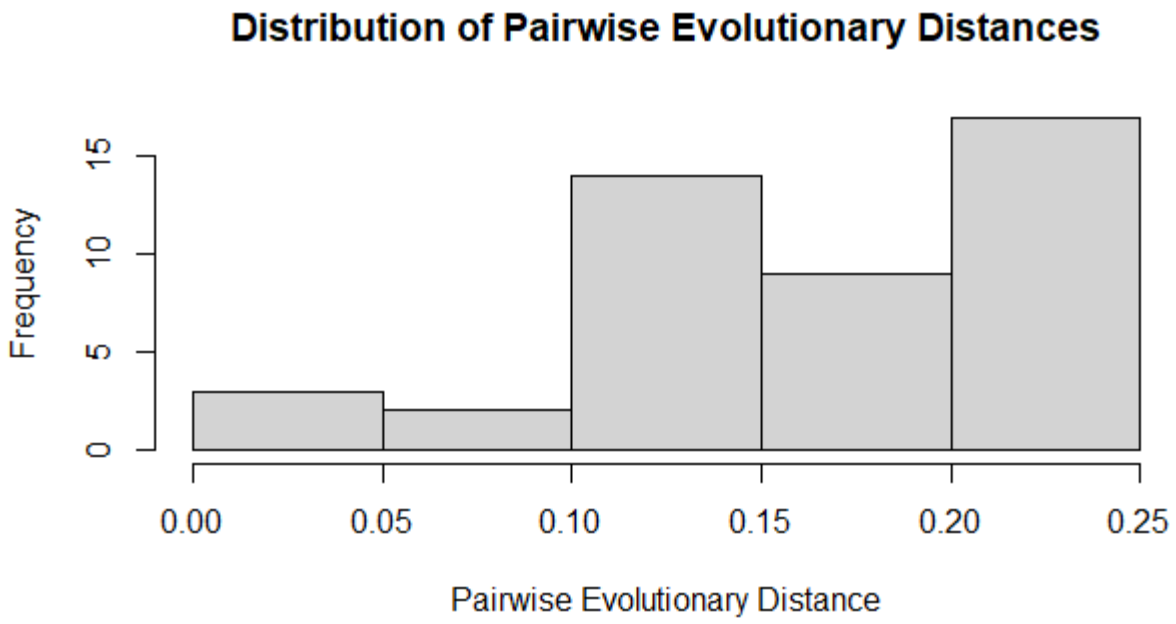
5.PAIRWISE EVOLUTIONARY DISTANCES

Pairwise evolutionary distances were calculated using the Poisson amino acid substitution model. The nonzero distances ranged from 0.0183 to 0.2472. The smallest recorded distance was 0.0183, observed between the chimpanzee and human sequences and between the human and rhesus macaque sequences. The largest distance was 0.2472 between the cow and mouse sequences. These values show different levels of sequence divergence among the selected mammals.

PAIRWISE EVOLUTIONARY DISTANCE HEATMAP



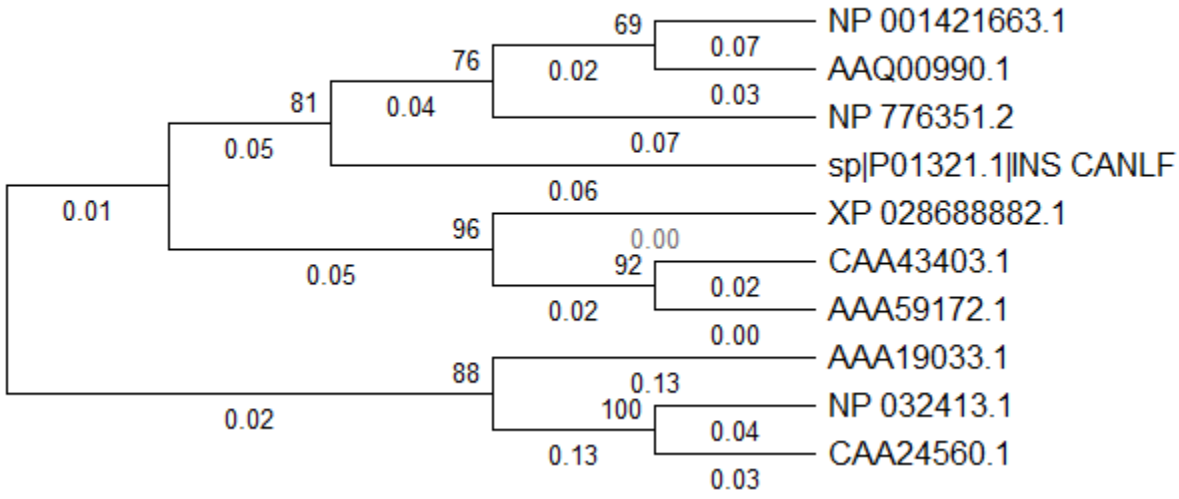
DISTRIBUTION OF PAIRWISE DISTANCES



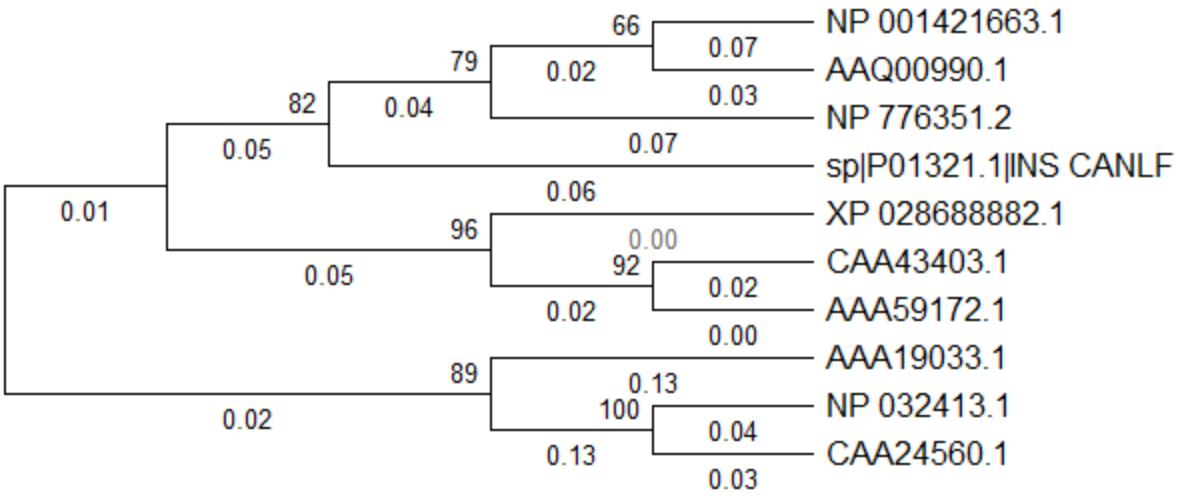
6.PHYLOGENETIC ANALYSIS

Neighbor Joining and Maximum Likelihood methods were used to construct phylogenetic trees. The recorded bootstrap values ranged from 69 to 100 for the Neighbor Joining analysis and from 66 to 100 for the Maximum Likelihood analysis. The trees provided two approaches for examining the relationships among the selected insulin protein sequences.

NEIGHBOR JOINING TREE

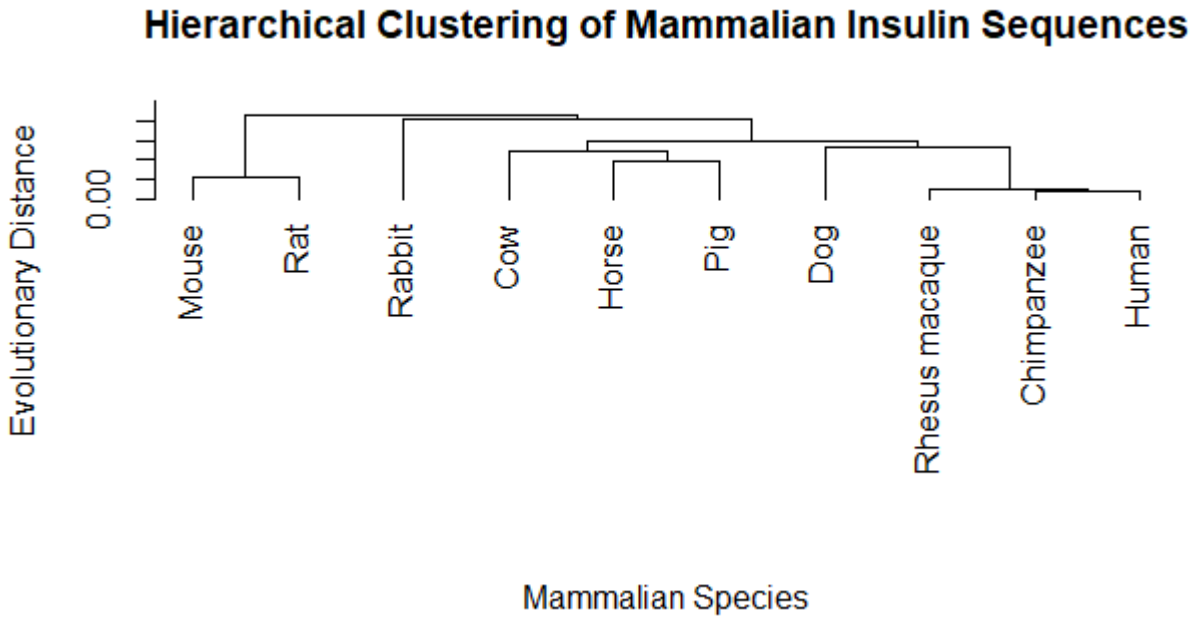


MAXIMUM LIKELIHOOD TREE



HIERARCHICAL CLUSTERING DENDROGRAM

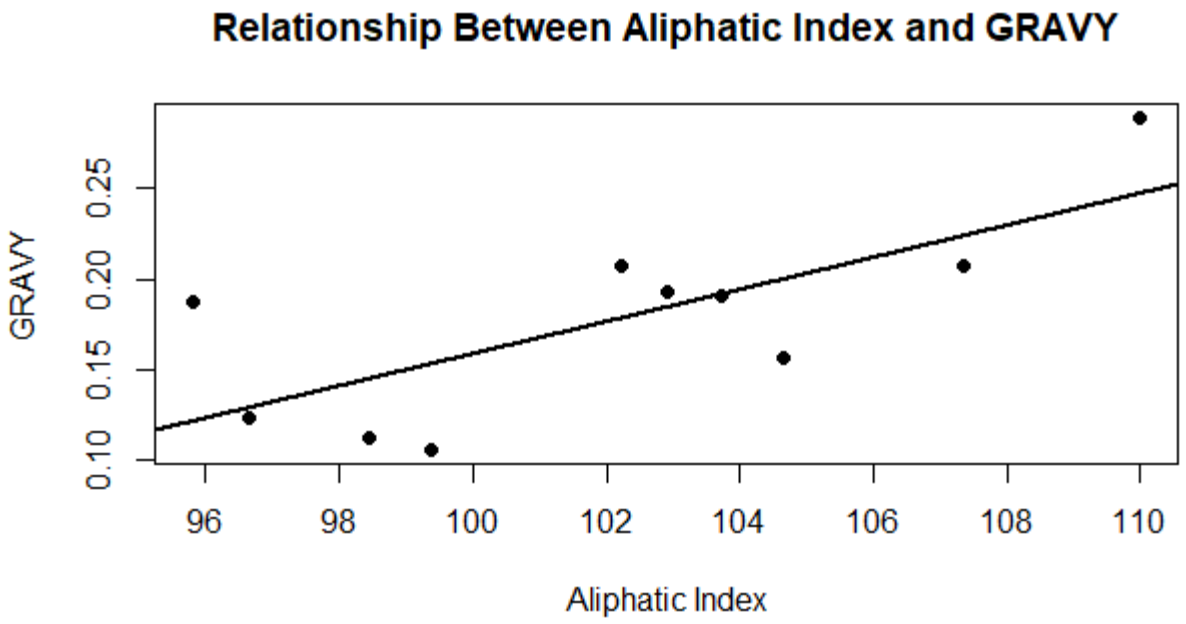
Hierarchical clustering dendrogram of insulin protein sequences from ten selected mammalian species based on pairwise evolutionary distances.



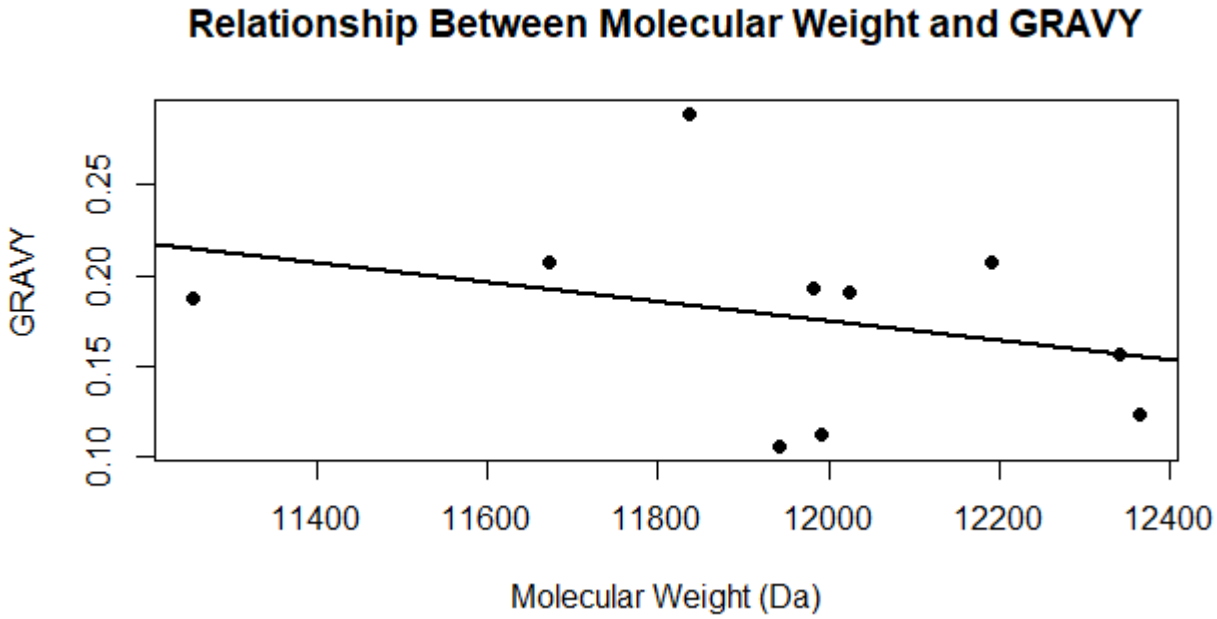
7.STATISTICAL ANALYSIS

Pearson correlation analysis showed a positive correlation between aliphatic index and GRAVY ($r = 0.7409$, $p = 0.01423$). This relationship was statistically significant at the 0.05 level. In contrast, molecular weight and GRAVY showed a negative correlation ($r = -0.3150$, $p = 0.3754$), which was not statistically significant.

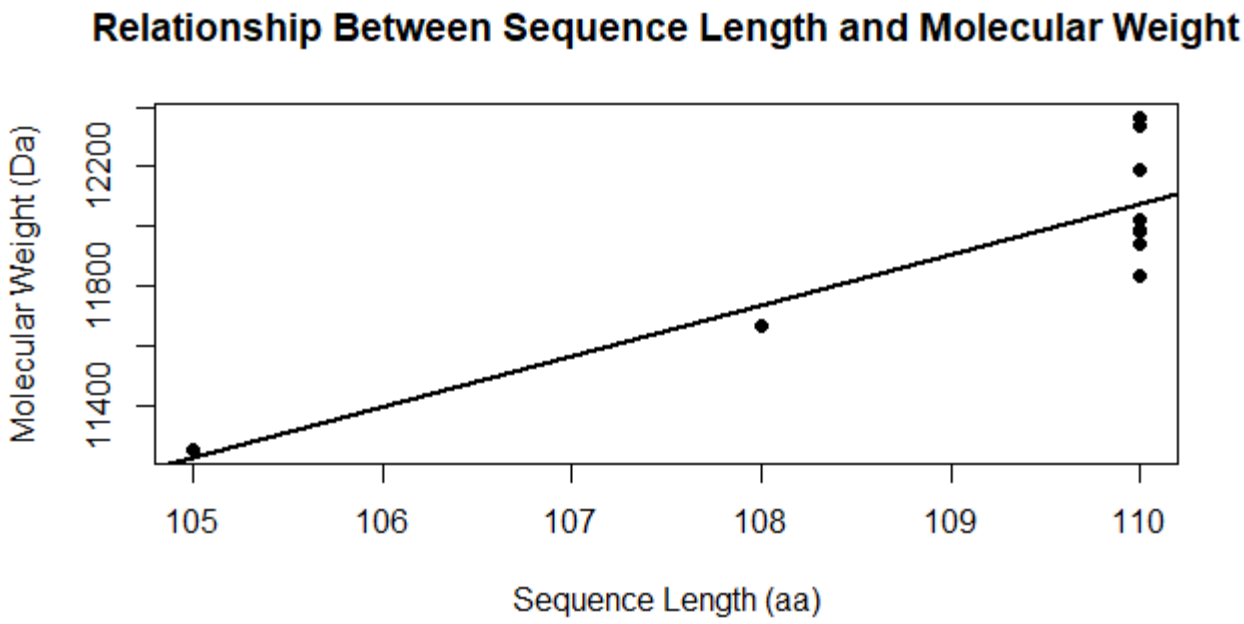
ALIPHATIC INDEX VS GRAVY



MOLECULAR WEIGHT VS GRAVY



MOLECULAR WEIGHT VS SEQUENCE LENGTH



8. GRAPHICAL REPRESENTATION

R software was used to visualize amino acid composition, physicochemical properties, evolutionary distance patterns, hierarchical clustering, and correlations between selected variables. The resulting figures provide graphical support for the numerical findings presented in this study.

5. DISCUSSION

The present study compared insulin protein sequences from ten selected mammalian species to investigate their amino acid composition, physicochemical properties, conserved and variable sites, and evolutionary relationships. The results revealed both similarities and differences among the selected sequences, providing information about insulin protein conservation across mammals.

The amino acid composition analysis showed that leucine was the most abundant amino acid, followed by glycine and alanine. These findings describe the overall composition of the analyzed sequences. Differences in amino acid percentages among species may reflect sequence variation, although the biological effects of individual substitutions cannot be established from composition data alone.

The physicochemical analysis revealed variation in molecular weight, theoretical isoelectric point, instability index, aliphatic index, and GRAVY values. These differences indicate that the selected sequences have different predicted physicochemical characteristics. However, computational predictions do not directly establish the actual stability, structure, or biological activity of insulin in living organisms.

Multiple sequence alignment identified 68 conserved sites and 42 variable sites in the recorded analysis. Conserved positions indicate that particular amino acids are shared across the aligned sequences, whereas variable positions reveal sequence differences among species. The presence of conserved positions is consistent

with the important biological role of insulin. Nevertheless, the functional significance of individual variable positions would require further investigation.

Pairwise, evolutionary distance analysis showed values ranging from 0.0183 to 0.2472. Smaller distances indicate greater sequence similarity, while larger distances indicate greater divergence under the selected model. The distance heatmap and hierarchical clustering dendrogram provided visual comparisons of sequence similarity. These patterns should be interpreted alongside the Neighbor Joining and Maximum Likelihood phylogenetic trees rather than treated as independent confirmation of evolutionary relationships.

The statistical analysis showed a positive correlation between aliphatic index and GRAVY ($r = 0.7409$, $p = 0.01423$). This suggests that, within the ten sequences examined, higher aliphatic index values tended to occur with higher GRAVY values. The relationship between molecular weight and GRAVY was negative but not statistically significant ($r = -0.3150$, $p = 0.3754$). Because the study included only ten sequences, these correlations should be interpreted cautiously and considered exploratory.

Overall, the combined sequence, physicochemical, distance, clustering, and phylogenetic analyses provided complementary information about similarities and differences among the selected mammalian insulin proteins. The findings support the use of comparative bioinformatics for investigating protein conservation and sequence variation. Further analyses using additional sequences and structural or functional evidence could provide a deeper understanding of the biological significance of the observed differences.

The conservation and variation observed in the insulin protein sequences provide insight into the evolutionary relationships among the selected mammalian species. Although insulin has an important biological role across vertebrates, its sequence and processing can differ during evolution. Comparative research on insulin and related proteins has also documented evolutionary diversity across organisms (Irwin, 2021; Chrudinová et al., 2026). These findings provide useful context for interpreting the conserved sites, variable sites, and pairwise evolutionary distances observed in the present study.

6.CONCLUSION

This study compared insulin protein sequences from ten selected mammalian species using bioinformatics tools. The analysis showed that insulin sequences share several conserved positions, indicating that important regions of the protein have been maintained across these species. The multiple sequence alignment identified 68 conserved sites and 42 variable sites across 110 alignment positions.

The physicochemical analysis revealed differences in molecular weight, theoretical isoelectric points, instability index, aliphatic index, and GRAVY values among the selected species. Pairwise evolutionary distances ranged from 0.0183 to 0.2472, showing different levels of sequence similarity. Phylogenetic analysis using Neighbor Joining and Maximum Likelihood methods provided further information about evolutionary relationships among the insulin sequences.

Statistical analysis showed a significant positive correlation between aliphatic index and GRAVY ($r = 0.741$, $p = 0.014$), whereas the correlation between molecular weight and grand average of hydropathicity GRAVY was not statistically significant ($r = -0.315$, $p = 0.375$).

Overall, this study demonstrates how bioinformatics tools can be used to investigate protein sequence conservation, physicochemical properties, and evolutionary relationships. These findings improve our understanding of insulin sequence variation among mammals. Further studies using additional species and larger datasets could provide a broader understanding of insulin evolution.

7.REFRENCES

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